



**JSC «Kazakh-British Technical University»
School of IT and Engineering**

COURSE PROJECT

Industrial Network and Fieldbuses

Research and practical implementation of data exchange between Modicon M241 PLC and ATV630 variable frequency drive via Modbus TCP with visualization on HMIST6400 HMI panel

**Checked
Completed by
the students**

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INTRODUCTION

A small pumping station provides a clear example of why an industrial network must connect operator commands, automatic control and equipment diagnostics. The operator needs to enter a pressure set point, start or stop the installation, and observe its condition. The controller must translate those requests into permitted actions, while the variable frequency drive controls the motor. A pressure measurement closes the control loop and makes the difference between a requested value and an actual process value visible.

The selected system consists of a Schneider Electric Modicon M241 PLC, one Altivar Process ATV630 variable frequency drive, a Harmony HMIST6400 HMI panel and a pressure transmitter. Ethernet with Modbus TCP provides the proposed communication path between the control devices. A TM3 analog input module connects the transmitter to the PLC. The study uses a single pump and an open water reservoir, avoiding the additional sequencing required by multiple pumps.

The central problem is to establish one clear source of drive commands. The ATV630 must remain stopped after power is applied and must start only when the PLC accepts a new command entered on the HMI. Manual mode changes the frequency reference; Auto mode regulates measured discharge pressure. Both modes preserve the same start permission, stop priority and fault handling.

The aim of this SIS1 study is to develop a documented technical basis for this system. Its objectives are to compare existing solutions, describe the pumping process, develop an original structural diagram, specify the equipment, verify its electrical and communication compatibility, and draw the pressure sensor connection. The resulting deliverables are a report and two editable Visio drawings. Network configuration, PLC programming, HMI development and commissioning are subsequent implementation activities; experimental performance is not claimed here.

1 Review of existing technical solutions

1.1 Ways to control a small pump

The simplest arrangement uses a pressure switch and a contactor to run a motor at its rated speed. Pressure is maintained within an on and off band rather than continuously adjusted. This arrangement can be adequate when demand is modest and frequent starts are acceptable, but an operator display receives little information unless additional instruments are installed. For this project it does not provide the required exchange of reference values and detailed drive condition over Modbus TCP.

A second arrangement uses a variable frequency drive with an internal pressure regulator. The pressure transmitter connects directly to the drive, which adjusts motor frequency to maintain its local set point. This reduces the work performed by the PLC and can suit a standalone pump. However, it moves the pressure control loop away from the M241. Using a PLC regulator and a drive regulator simultaneously would create two competing control actions and complicate fault diagnosis.

A third arrangement places the regulator, operating mode selection and process permissions in the PLC. The drive receives a run command and frequency reference, while the HMI exchanges requests and displayed values with the PLC. This is the selected architecture because the course topic concerns data exchange and because the initial condition assigns control authority to the PLC. The controller can associate each command with the measured pressure, the state of the pump and the validity of communication.

Selection must be based on the required behavior rather than the number of functions advertised by a product. A suitable system needs a pressure input, an adjustable motor frequency, an operator interface and a defined response to lost data. These requirements form the basis of the comparison in Table 1.

1.2 Comparison of control architectures

Table 1 compares three practical arrangements for the proposed single pump installation.

Table 1 - Comparison of existing control arrangements

Criterion	Contactor and switch	Drive regulator	PLC regulator
Pressure control	On and off band	Continuous in drive	Continuous in PLC
Operator reference	Local switch setting	Drive or HMI setting	HMI request to PLC
Drive information	No drive installed	Available from drive	Collected by PLC
PLC command authority	Limited sequencing	Depends on settings	Explicitly centralized
Fit to this topic	Low	Possible alternative	Selected arrangement

A hardwired 0–10 V or 4–20 mA frequency reference is another interface option. Separate wires are needed for run permission and selected status signals. Detailed faults, current and power require additional channels or a communication link, so an analog reference does not remove the need to plan diagnostic exchange.

Modbus RTU over RS-485 can carry commands and measured values in a compact serial installation. Modbus TCP instead uses Ethernet infrastructure and supports the proposed HMI, PLC and drive arrangement through a switch. Ethernet describes the network technology; Modbus TCP describes the application messages used over TCP/IP. They are complementary parts of the same selected solution [2].

For one drive, a switched star makes each connection easy to identify and provides access for an engineering computer. Network redundancy and multi-pump sequencing are outside this installation. The selected small network keeps the pressure loop and communication roles easy to document.

1.3 Equipment vendors and communication choices

Schneider Electric offers the selected M241, ATV630 and Harmony ST6 families. M241 includes Ethernet and supports TM3 expansion; HMIST6400 supplies the display; ATV630 provides the motor power stage. This manufacturer ecosystem simplifies documentation collection. PLC development uses EcoStruxure Machine Expert, and HMI development uses EcoStruxure Operator Terminal Expert [1, 2, 4].

ABB provides an alternative communication arrangement through its FMBT-21 Modbus TCP drive adapter. Its January 2023 guide describes fieldbus command and reference selection, network settings, and separate control and status words. This demonstrates that a drive can support the required application protocol through a selected adapter. A replacement would require confirmation of drive compatibility, the adapter, the control profile and register interpretation, even though the application remains pressure control [9].

Siemens offers SINAMICS G120X converters for water, wastewater and building services. The July 2024 catalog distinguishes versions with PROFINET and EtherNet/IP from versions with Modbus RTU and other serial protocols. This is an important selection distinction: the presence of an Ethernet connector does not establish support for Modbus TCP. A Siemens alternative would need a verified protocol path and corresponding controller configuration [10].

Pressure instrument manufacturers also offer different signal variants of the same sensor family. WIKA A-10 is selected in its two-wire 4–20 mA version, rather than a voltage output version. Its 0–10 bar gauge range covers the proposed pressure operating window. The order specification includes the water stability option identified by the current data sheet for this range [7].

The Schneider arrangement fits the stated course topic, HMIST6400 and one ATV630. Its measurement path can be checked against manufacturer documents. This project-specific selection does not establish a universal cost or performance advantage over other vendors.

1.4 Data exchange and selection rationale

In the proposed communication model, the M241 acts as a Modbus TCP client when exchanging data with the ATV630 and as a server for HMI access to application variables. The HMI requests or reads PLC data; it does not write run commands directly into the drive. The M241 programming guide describes its client and server services, providing the technical basis for this arrangement [2].

Drive commands include the permitted run state and frequency request. Feedback includes operating state, actual frequency, faults and electrical values. Pump pressure comes from PT01 through TM3. Process pressure and motor frequency must therefore remain separate quantities in the software and display.

The ATV600 programming documentation describes command and reference channels, operating profiles and diagnostics. The drive channel must be assigned to the selected network interface so that a local keypad or a terminal command cannot unexpectedly take authority. Schneider guidance also identifies embedded Ethernet as the source for network reference control [5, 6]. The precise control profile and register mapping will be selected consistently during implementation.

Register numbers alone are insufficient for a correct data model. A later mapping must identify the manufacturer parameter, protocol address convention, data type, sign, scaling and unit. An unsigned raw value representing tenths of hertz must not be displayed as a pressure value. Multi-register values require a confirmed word order. Commands should be written through dedicated PLC logic, while monitoring values should be exposed as read-only HMI objects.

The selected solution combines an analog pressure input, a PLC regulator, one Ethernet drive and one HMI. A communication watchdog and explicit restart policy complete the architecture. These requirements form the basis of later implementation and testing; they do not establish commissioned performance.

2 Description of the technological process

2.1 Single pump pressure control

The proposed process is a small recirculating water installation. Water leaves an open reservoir T01, passes through centrifugal pump P01, and returns through an adjustable valve V01. The valve represents changing demand or pipeline resistance. Pressure transmitter PT01 measures discharge pressure upstream of the valve. A mechanical relief branch RV01 returns water to the reservoir if pressure reaches its selected relief setting.

Motor M01 drives the pump and receives variable frequency power from the ATV630. The M241 reads pressure, evaluates operating permissions and determines the frequency request. HMIST6400 shows pressure SP and PV, operating mode and drive condition. The adjustable valve is operated by hand and does not require a motorized actuator or another PLC output.

The proposed normal pressure set point range is 1.0–3.0 bar gauge, with 2.0 bar as an initial design value. The active frequency range is initially limited to 20–50 Hz; Stop disables running rather than requesting the minimum running frequency. A process high pressure trip is proposed at 4.5 bar and a separate relief setting at 5.0 bar. These values are project design assumptions, subject to the selected pump curve and the ratings of all wetted components.

The intended duty point for pump selection is approximately 1 m³/h at 3 bar. A 0.75 kW motor class is used for the electrical study, but hydraulic suitability cannot be inferred from motor power alone. A later pump selection must verify its head and flow curve, minimum permitted speed, suction conditions and operation with the discharge restriction. The reservoir low level contact LSL01 inhibits running to reduce the risk of dry operation.

This process keeps the main control problem observable: changing V01 changes the load, PT01 measures the result, and the PLC adjusts motor frequency in Auto. No flow meter, second pump or cascade control is required for SIS1.

2.2 Structural diagram and signal paths

Figure 1 presents the original structural diagram of the pumping process and its control system. Solid process lines show water flow, the dashed line shows analog feedback, and the labeled network connections carry Modbus TCP.

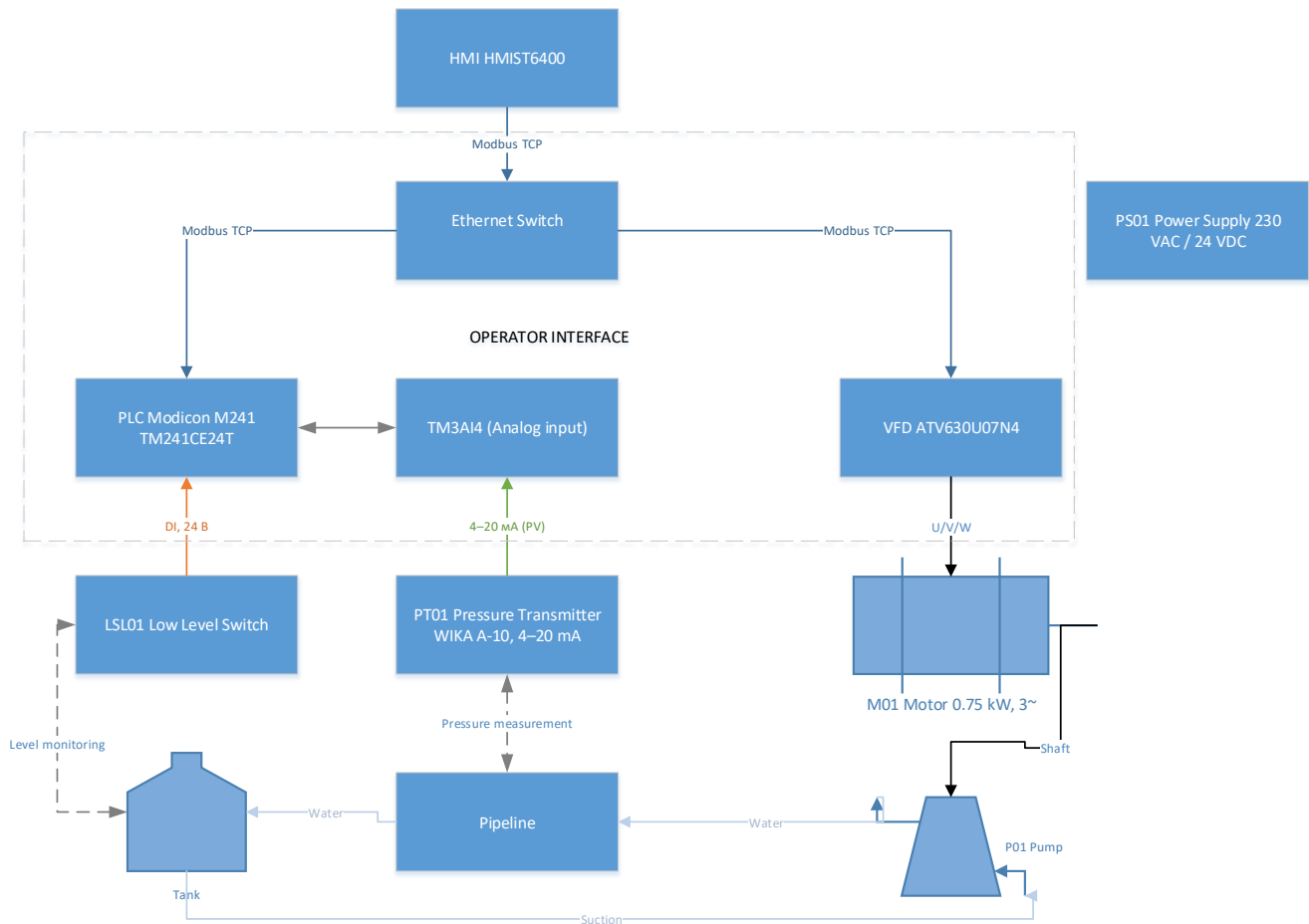


Figure 1 - Structural diagram of the single pump pressure control system

PT01 supplies pressure feedback to TM3AI4; the TM3 expansion bus transfers the input to M241. The Ethernet switch joins the HMI, PLC and drive. The drive supplies the motor, and the motor transfers mechanical power to the pump. Table 2 identifies the main diagram elements and their functions.

2.3 Main elements and operating sequence

Table 2 - Main elements of the structural diagram

Tag	Element	Function
A01	Modicon M241	Control logic and command authority
A02	TM3AI4	Pressure current input to PLC
A03	HMIST6400	Operator commands and visualization
A04	ATV630	Motor frequency and power control
SW01	Ethernet switch	Connect HMI, PLC and drive
PT01	Pressure transmitter	Measure discharge pressure PV
LSL01	Low level contact	Inhibit pump at low reservoir level
M01 / P01	Motor and pump	Transfer water and create pressure
T01 / V01	Reservoir and valve	Supply water and vary resistance
RV01	Relief valve	Limit pressure mechanically

At power-up, the application clears the run request and initializes a stopped state. Selection of Manual or Auto does not itself initiate motion. The PLC first requires valid pressure input, healthy communication, sufficient reservoir level and a drive ready condition. Only a new HMI Start request can set the run request after these conditions are satisfied.

In Manual, the operator enters a frequency request within the permitted limits. In Auto, the operator enters pressure SP and the PLC regulator determines frequency from SP minus PV. Stop takes priority in both modes. A mode change is initially allowed only while stopped, which avoids an unplanned jump between manually entered frequency and regulator output.

2.4 Operating variables and supervision

Table 3 defines the required variables by meaning and direction. The pressure SP and PV are distinct from the drive frequency request and feedback.

Table 3 - Required operator and monitoring variables

Variable	Source and direction	Meaning and unit
Manual / Auto	HMI request → PLC	Mutually exclusive mode selection
Start / Stop	HMI request → PLC	Run request / stop with priority
SP	HMI → PLC	Pressure reference, 1–3 bar
PV	PT01 → PLC → HMI	Measured gauge pressure, bar
Manual frequency	HMI → PLC → drive	Permitted request, 20–50 Hz
Status	Drive / PLC → HMI	Stopped, ready, running or faulted
Health	PLC → HMI	Data freshness and valid permissions
Fault	Drive / PLC → HMI	Code and cause requiring attention
Current	Drive → PLC → HMI	Motor output current, A
Voltage / Power	Drive → PLC → HMI	Selected motor values, V / kW

Health is a PLC-defined summary, not an assumed manufacturer register. It combines communication validity and relevant process permissions. If data become stale, the HMI should mark the affected values unavailable and show the communication fault. A drive fault, an invalid pressure signal, low reservoir level or excessive pressure clears the run request. Restoring the condition or resetting a fault does not start the motor; a new Start request is required.

3 Equipment specification

3.1 Control and communication equipment

Table 4 specifies the control equipment. TM241CE24T is the proposed Ethernet-capable M241 reference; it is a selection for this design and must be checked against the controller actually available in the laboratory.

Table 4 - Control equipment specification

Equipment	Selected reference	Principal parameters
PLC A01	TM241CE24T	24 V DC; 14 DI; 10 PNP DO; one Ethernet port [1]
Analog module A02	TM3AI4	4 AI; 4–20 mA mode; 12-bit conversion; external 24 V DC [3]
HMI A03	HMIST6400	7-inch; 800 × 480; two Ethernet ports; 24 V DC $\pm 20\%$ [4]
Drive A04	ATV630U07N4	0.75 kW normal duty; 380–480 V 3-phase input; 2.2 A output; embedded Modbus TCP [5, 8]
Switch SW01	Industrial 5-port type	RJ45 10/100 Mbit/s; 24 V DC; ≥ 3 active ports
Supply PS01	24 V DC regulated	Proposed 5 A class; protected branches; size for total load and inrush

The PLC, analog module, HMI and switch use low voltage control power. The drive uses a separate three-phase mains supply. Motor power must never pass through PLC outputs. The power supply current rating and protective devices are finalized from the selected component manuals, wiring conditions and simultaneous loads; HMI and PLC startup inrush must also be considered.

A switch is specified by required characteristics rather than an unverified part number. Short shielded Ethernet patch cables and cabinet separation between motor power wiring and measurement wiring support a clear, maintainable installation.

3.2 Sensors and process equipment

Table 5 separates catalog-backed selections from equipment requirements that depend on the final hydraulic installation.

Table 5 - Sensor and actuator specification

Element	Selection or design requirement	Electrical or process range
PT01	WIKA A-10; gauge; G $\frac{1}{4}$ A; DIN form A; water stability option [7]	0–10 bar; two-wire 4–20 mA; 8–30 V DC
LSL01	Float switch with dry contact	Binary low level indication; 24 V DC input circuit
M01	VFD-compatible 3-phase motor	0.75 kW class; 400 V; 50 Hz; nameplate current ≤ 2.2 A
P01	Centrifugal water pump	Duty target 1 m ³ /h at 3 bar; curve and speed range to be verified
T01 / V01	Open reservoir / manual valve	Water service; pressure-bearing parts rated ≥ 10 bar
RV01	Mechanical relief valve	Proposed setting 5 bar; capacity sized for pump flow

The pressure transmitter is the continuous measurement device. Its DIN form A connection is selected so that the electrical diagram can identify the positive and negative terminals unambiguously. The float switch is a separate discrete permission and does not measure a continuous level.

The ATV630 converts fixed-frequency mains power into variable-frequency motor power. The motor and pump form the process actuator. At 400 V, the selected drive belongs to the appropriate supply-voltage class, while the motor current must remain within the drive rating after applicable derating. A practical motor or pump model is not certified as suitable until its nameplate and pump curve have been reviewed.

4 PLC compatibility

4.1 Analog and discrete input compatibility

The pressure transmitter produces 4–20 mA and TM3AI4 supports that current input range. The module provides an input measuring circuit, but not the transmitter loop supply. An external 24 V DC source is therefore included. The manufacturer hardware guide gives a maximum current-input impedance of 50 Ω and identifies the I0+ and I0– channel terminals used in Figure 2 [3]. External loop power is also confirmed in Schneider support guidance [11].

The transmitter load limit is $R_{load} \leq (U_{supply} - 8 \text{ V}) / 0.020 \text{ A}$ [7]. At a nominal 24 V supply, this gives 800 Ω . A 50 Ω module input consumes at most 1 V at 20 mA, leaving substantial nominal loop-voltage margin for the transmitter and wiring. At the proposed minimum control supply of 20.4 V, the load limit is 620 Ω . Compatibility must be checked using total loop resistance, including terminals, cables and any added protection.

For a 0–10 bar range, engineering pressure is calculated as $PV = 10 \times (I - 4) / 16$, where I is expressed in mA. Thus 4 mA corresponds to 0 bar, 12 mA to 5 bar and 20 mA to 10 bar. The application must use the actual channel scaling configuration rather than assume a raw numerical range. A design-invalid signal below 3.6 mA or above 21 mA is treated as a process fault; these are proposed validity thresholds, not a claim of certified transmitter alarm behavior.

LSL01 is wired as a dry contact in a 24 V DC digital input circuit. A contact indicating sufficient level is required for running; an open circuit inhibits the pump. The common terminal and input polarity must follow the actual M241 sink/source wiring configuration. This arrangement detects loss of permission, but does not establish a safety integrity rating.

4.2 Ethernet and drive compatibility

The Ethernet-capable M241, HMIST6400 and ATV630 are assigned distinct addresses on one isolated subnet. Table 6 gives the proposed address plan. It is a project allocation, not a record of applied settings.

Table 6 - Proposed Ethernet address plan

Device	IP address	Role
M241	192.168.10.10	Client to drive; server to HMI
ATV630	192.168.10.20	Modbus TCP server
HMIST6400	192.168.10.30	Client to PLC
Engineering PC	192.168.10.100	Temporary configuration access

The proposed subnet mask is 255.255.255.0. No router is required within this subnet; unused gateway fields are configured according to each device manual. Modbus TCP uses TCP port 502 [2]. The unused HMI Ethernet port is not assumed to be a transparent switch. All three control devices connect to SW01.

A later implementation must select one supported drive command profile and use the same profile when interpreting status. A proposed 100 ms drive exchange period and 500 ms HMI refresh period are initial design values, with a 1 s loss-of-communication criterion to be verified. The drive also needs its own communication-loss stop response so that PLC or cable failure cannot leave a previous run command active [5].

Motor current, motor output voltage and output power should be named and scaled according to the selected ATV630 parameters; estimated drive values must not be represented as independent calibrated measurements. The control network is kept separate from unrelated traffic. A normal HMI Stop remains an operational command. Emergency stopping requires a separately assessed hardwired safety arrangement; it is not performed through ordinary Modbus TCP or a standard M241 application.

5 Electrical connection diagram for the sensor

Figure 2 presents the connection of the two-wire PT01 transmitter to TM3AI4 channel 0. The DIN form A connector uses pin 1 for U+ and pin 2 for U- [7]. The module supply branch includes the 100 mA type T fuse shown in the TM3 hardware guide [3].

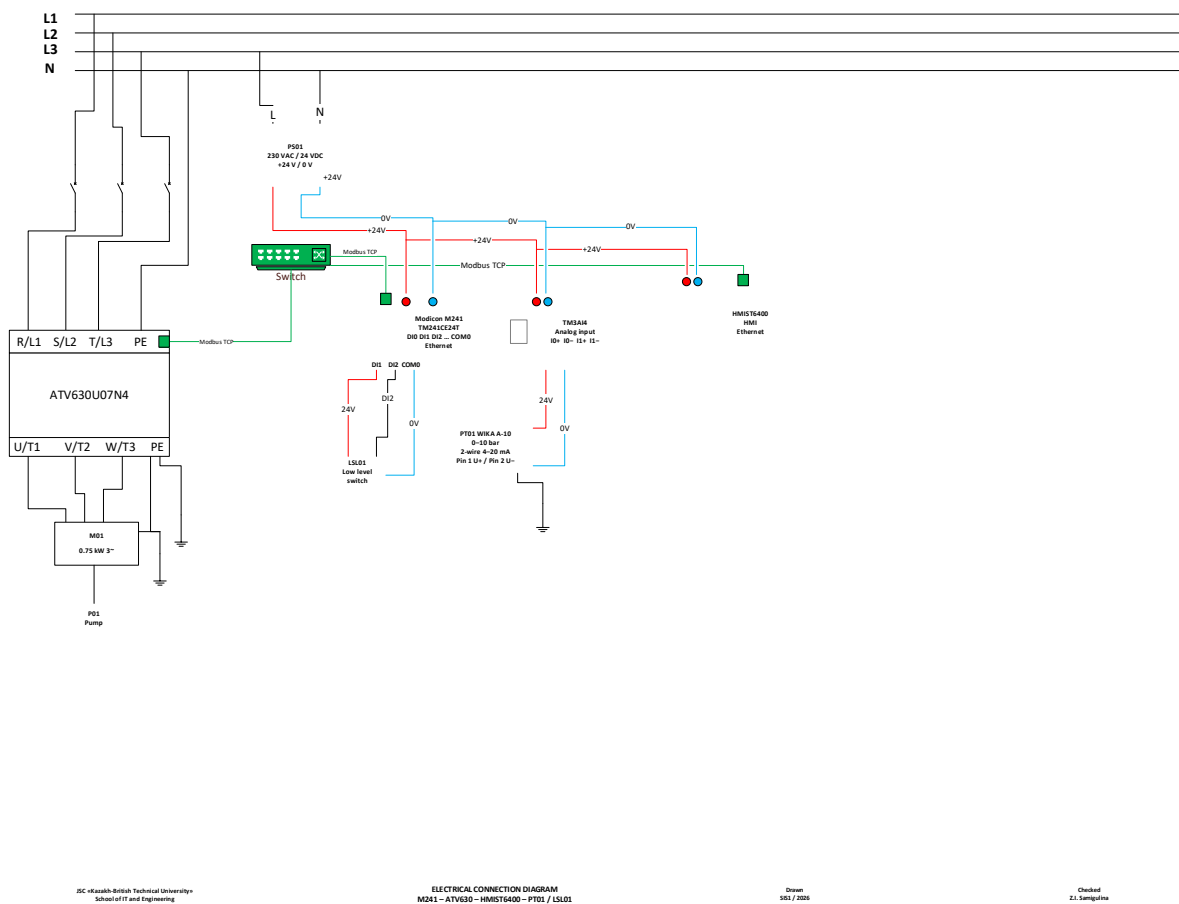


Figure 2 - Electrical connection of PT01 to TM3AI4 analog input channel 0

The current path is +24 V → PT01 pin 1 → PT01 pin 2 → I0+ → measuring input → I0- → 0 V. Module terminals 24V and 0V have a separate supply connection. The shield is bonded at the cabinet functional earth clamp; it is not a signal-return conductor. Unused and N.C. terminals remain unwired.

CONCLUSION

The SIS1 study develops a single pump pressure control system using Modicon M241, one ATV630 and the HMIST6400 panel. The comparison of control arrangements supports a PLC-centered architecture in which the HMI submits requests and the PLC retains authority over the drive. The proposed process gives SP and PV a clear physical meaning as requested and measured discharge pressure.

The equipment specification identifies an Ethernet-capable M241 reference, an analog input expansion module, an ATV630 voltage and current class, and a two-wire pressure transmitter. The compatibility analysis confirms the selected signal ranges and the need for external loop power. It also distinguishes the drive rating from the motor and pump requirements that must be checked during equipment selection.

Two original drawings describe the process structure and the pressure sensor wiring. The operating requirements preserve a stopped initial state, require an explicit HMI Start request, and prevent an automatic restart after a fault or communication recovery. These results provide the basis for later network configuration, control software development and practical verification. No laboratory commissioning or measured control performance is asserted in this stage.

LITERATURE

[1] Schneider Electric. Modicon M241 Logic Controller Hardware Guide. EIO0000003083.08, April 2026. [Manufacturer source](#) (accessed 1 October 2026).

[2] Schneider Electric. Modicon M241 Logic Controller Programming Guide. EIO0000003059.10, April 2026. [Manufacturer source](#) (accessed 1 October 2026).

[3] Schneider Electric. Modicon TM3 Analog I/O Modules Hardware Guide. EIO0000003131.04, July 2024, pp. 43–47. [Manufacturer source](#) (accessed 1 October 2026).

[4] Schneider Electric. Harmony ST6 User Manual. EIO0000003527.04, January 2024. [Manufacturer source](#) (accessed 1 October 2026).

[5] Schneider Electric. Altivar Process ATV600 Programming Manual. EAV64318.14, May 2026. [Manufacturer source](#) (accessed 1 October 2026).

[6] Schneider Electric. Basic parameters for embedded Modbus Ethernet communication on Altivar Process drives. FAQ FA300078, updated 17 April 2026. [Manufacturer source](#) (accessed 1 October 2026).

LITERATURE

[7] WIKA. Pressure transmitter for general industrial applications Model A-10. Data sheet PE 81.60, February 2026, pp. 3, 6 and 8. [Manufacturer source](#) (accessed 1 October 2026).

[8] Schneider Electric. AC Drives and Soft Starters Section 18 Digest 179. Catalog 0100CT2401-SEC-18, version 1.2, June 2025, p. 18-16. [Manufacturer source](#) (accessed 1 October 2026).

[9] ABB. FMBT-21 Modbus TCP adapter module Quick installation and start-up guide. 3AXD50000158560 Rev B, January 2023. [Manufacturer source](#) (accessed 1 October 2026).

[10] Siemens. SINAMICS G120X infrastructure converters for HVAC Water Wastewater. Catalog D 31.5, July 2024, technical specifications. [Manufacturer source](#) (accessed 1 October 2026).

[11] Schneider Electric. External loop power for TM3 analog current inputs. FAQ000240350, published 6 July 2022, updated 15 May 2026. [Manufacturer source](#) (accessed 1 October 2026).